

Juhyun Bae Research Portfolio



Juhyun Bae
배주현

M.S. Candidate
Department of Electrical Engineering
Korea University

Deep Image Processing Lab

Research Interests

- Image Restoration
- Computational Photography
- Data Augmentation
- Deep Learning

Honors & Awards

- 2nd Place, Samsung AI Challenge 2024
- Honorable Mention, Hanium ICT Mentoring Project Contest 2022

Research & Publications

- Trajectory-aware Differentiable Blur Augmentation for Image Deblurring (Target Venue: ICLR 2027, Ongoing)
- DnSAD: Defocus and Spatially Adaptive Deblurring for Display Mura Compensation, Journal of the Society for Information Display, 2026 (Co-first Author)
- Intrinsic Decomposition-based Low-Light Image Enhancement, IEEE Signal Processing Letters, 2025

Selected Projects

- Vision-based Safety Equipment Inspection (KIST AI-Robot Institute, 2023)
- Samsung AI Challenge, Machine Learning Force Fields (2024)
- Image Aesthetics Assessment using Vision-Language Models (2024)

Education

Korea University

M.S. Candidate in Electrical Engineering
(Advisor: Prof. Seung-Won Jung)
Feb, 2025 – Present

Seoul National University of Science and Technology

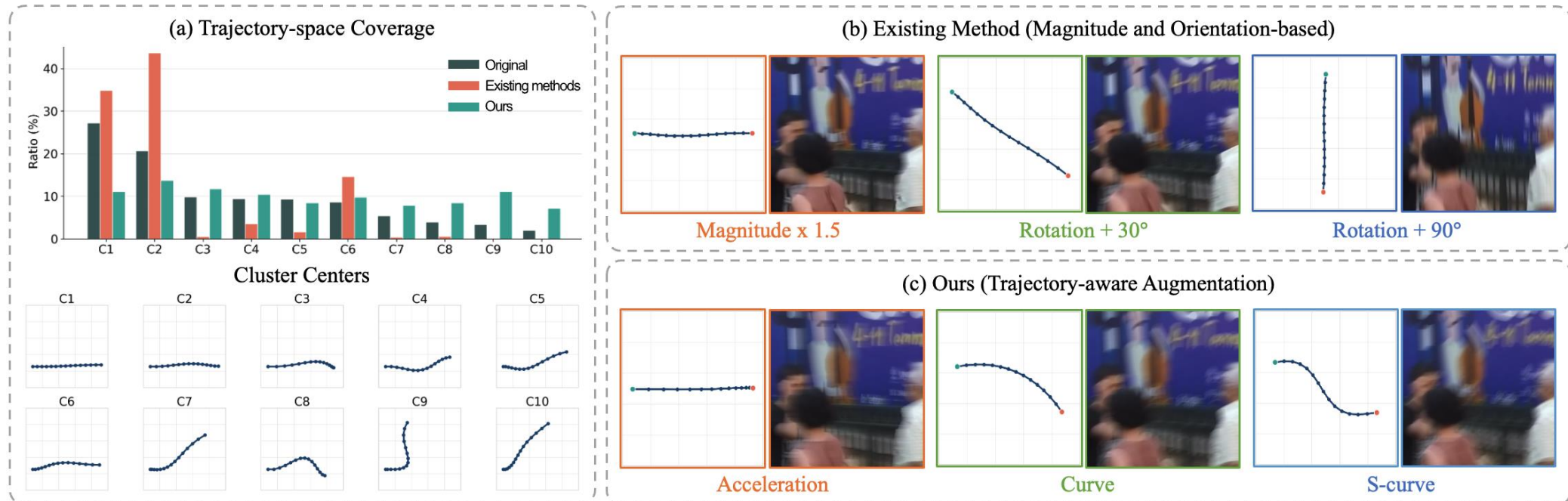
B.S. in Computer Science and Engineering
GPA: 4.38 / 4.50
Graduated as Valedictorian
Mar, 2020 - Feb, 2025

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

Target Venue: ICLR 2027

Problem & Motivation

- Motion blur is caused by camera shake and object motion during image exposure.
- Blur augmentation synthesizes additional blur from sharp images to expand deblurring training data.
- Existing blur augmentation varies only motion magnitude and orientation, limiting trajectory diversity.
- Trajectory-aware blur augmentation is proposed to enrich the motion trajectory space.



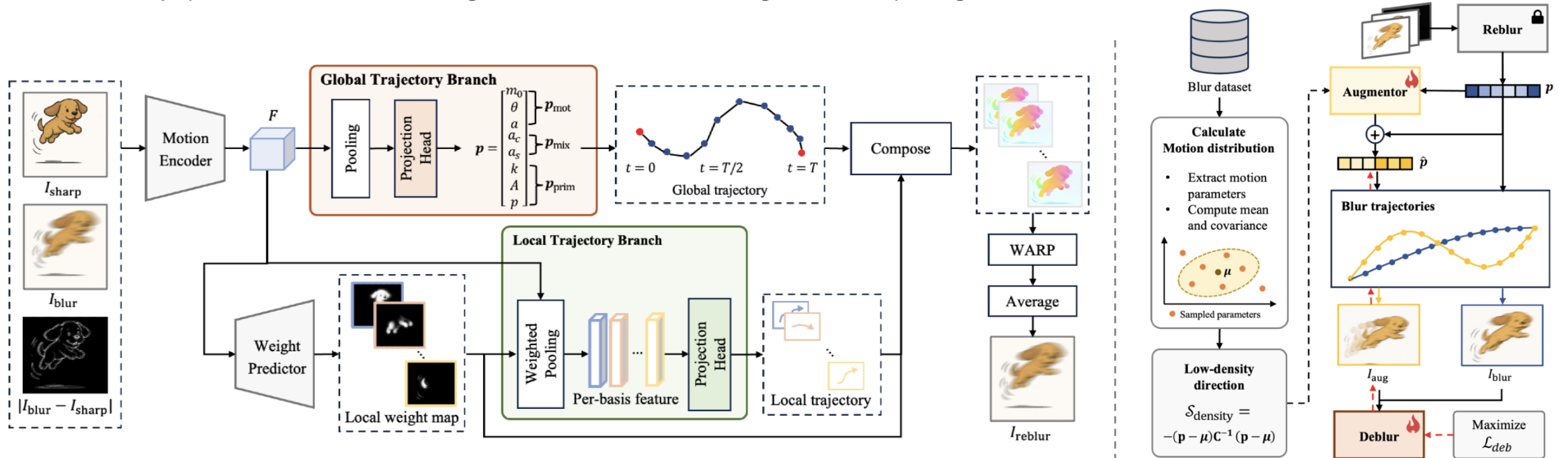
Comparison of trajectory diversity between existing blur augmentation and the proposed method.

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

Target Venue: ICLR 2027

Methods

- Train a trajectory-aware reblur network to model diverse motion trajectories, and utilize it for differentiable blur augmentation during deblurring training.
- Jointly model global camera motion and local object motion to generate realistic spatially varying motion blur.
- Jointly optimize learnable motion augmentation with the deblurring model to improve generalization.



Overview of the proposed trajectory-aware blur augmentation framework

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

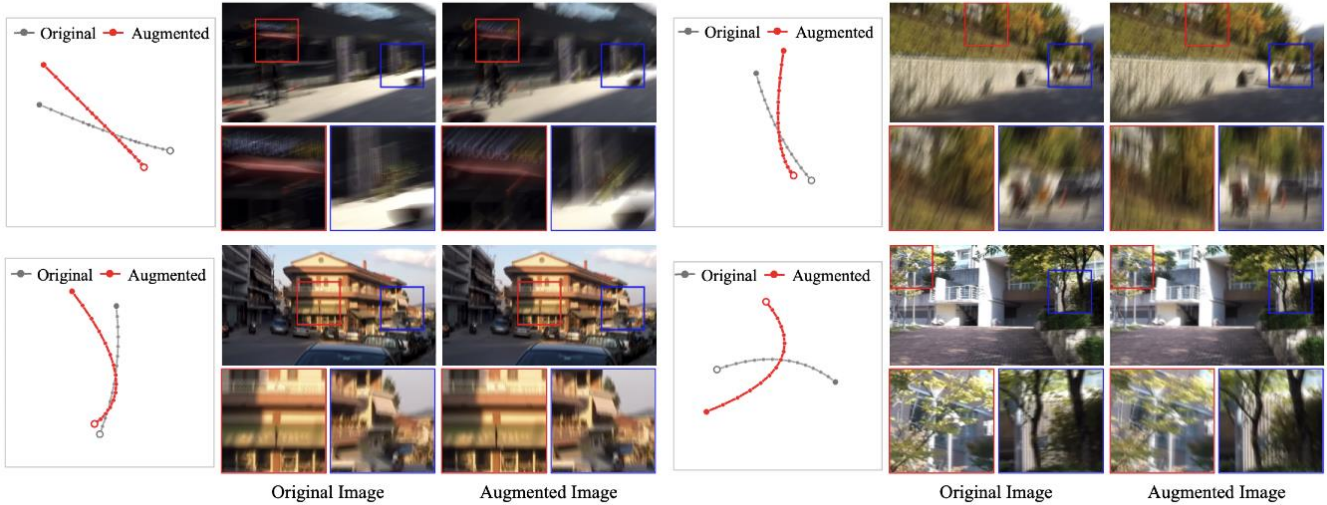
Target Venue: ICLR 2027

Results

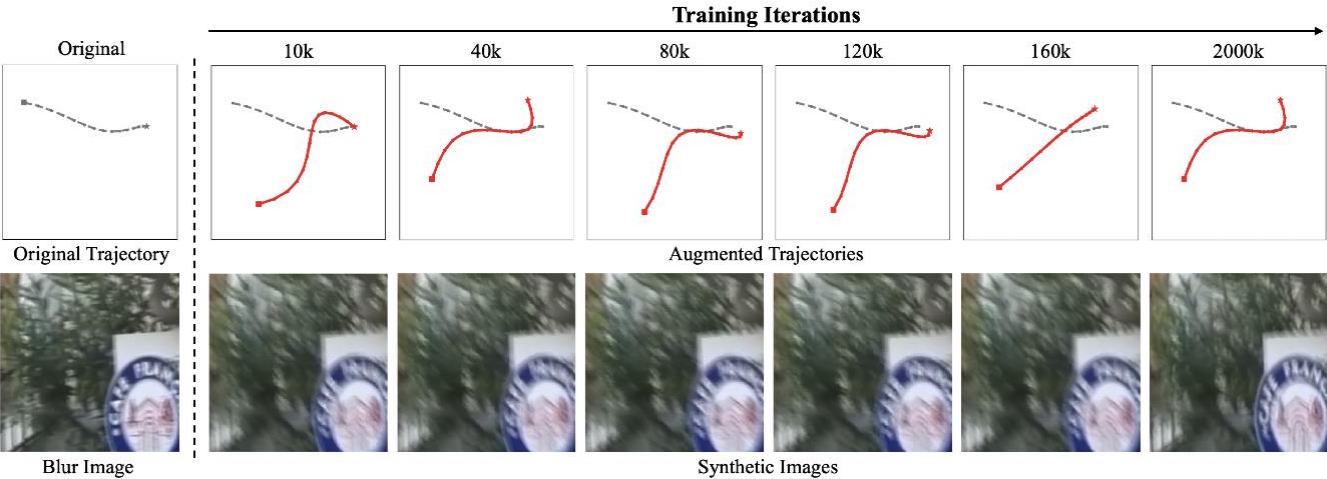
- +2.45 dB over baseline (GoPro 10%)
- Improved performance on HIDE benchmark

Model	GOPRO 10 %		GOPRO 100%	
	PSNR	SSIM	PSNR	SSIM
MIMO-UNet+	28.92	0.915	32.44	0.957
+ ID-Blau	30.61	0.940	32.93	0.961
+ Ours	31.37	0.947	33.10	0.962
HIDE (Test only)				
MIMO-UNet+	26.71	0.877	30.00	0.930
+ ID-Blau	28.50	0.910	30.68	0.938
+ Ours	28.98	0.919	30.71	0.938

Performance comparison of the proposed model and baseline methods.



Visualization of trajectory-aware augmentation and the corresponding blur variations.



Evolution of learnable trajectory augmentation during deblurring training.

Defocus and Spatially Adaptive Deblurring for Display Mura Compensation

Journal of the Society for Information Display, 2026 (Co-first author)

Project

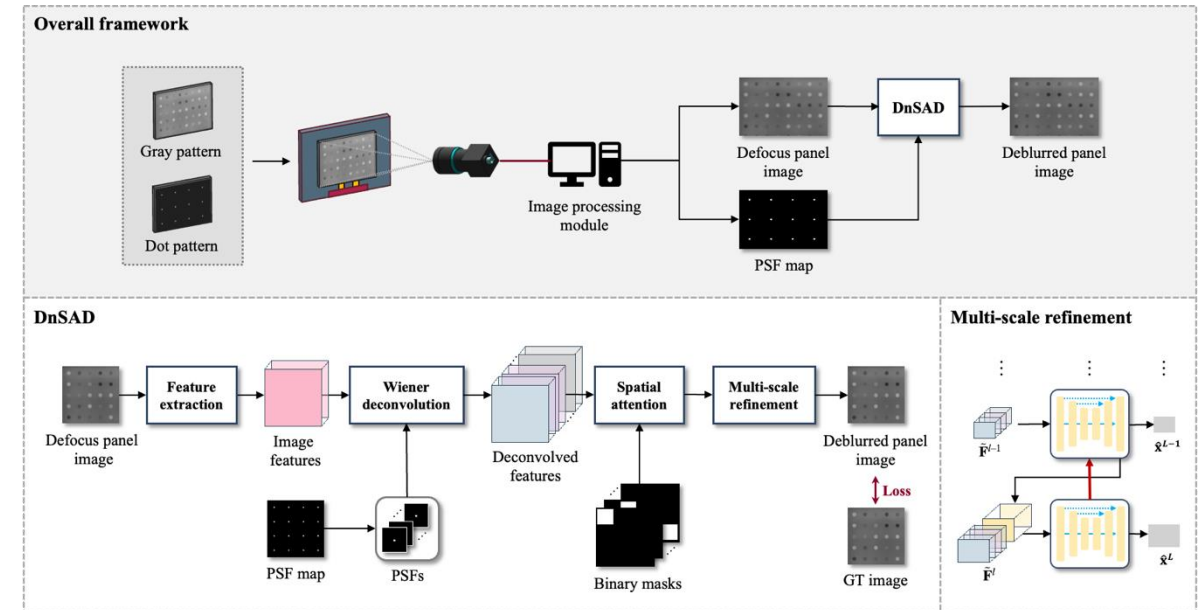
- Developed DnSAD as part of an LG Display industry-academia project to restore spatially varying defocus blur for display mura compensation.

My Contributions

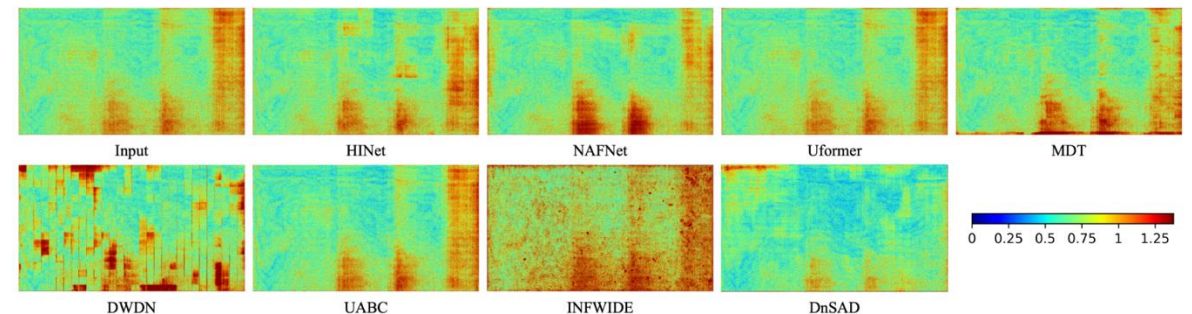
- Designed a PSF-aware deblurring framework leveraging measured PSF maps for spatially varying blur restoration.
- Introduced spatial attention and multi-scale refinement to effectively restore high-frequency mura patterns.
- Constructed a real-world OLED panel dataset and validated the framework for LG Display mura compensation.

Results

- Achieved 36.41 dB PSNR and 0.6474 MAE-MURA on real-world OLED panel datasets.
- Outperformed existing blind and non-blind deblurring methods.



Overview of the proposed PSF-aware deblurring framework (DnSAD).



Quantitative Results on Display Mura Compensation

Machine Learning Force Fields

Samsung AI Challenge 2024 – 2nd place

Project

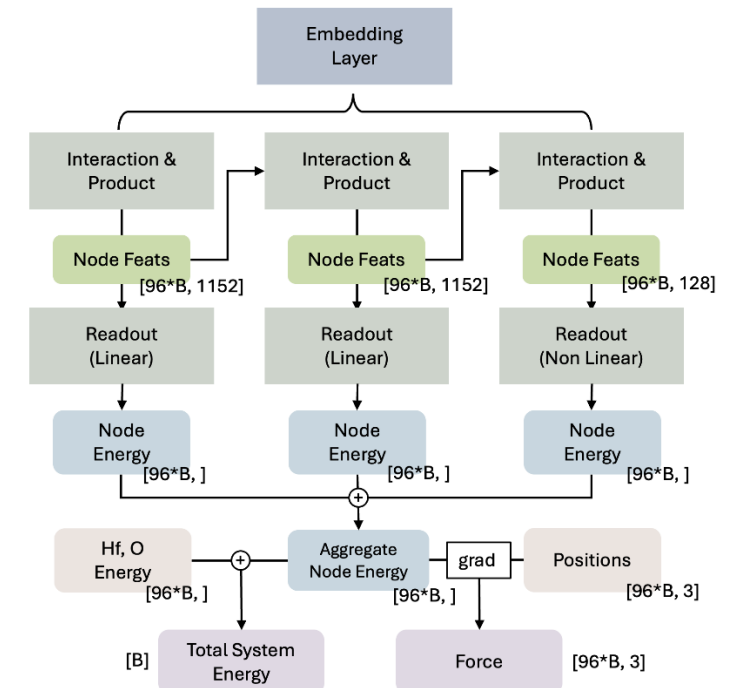
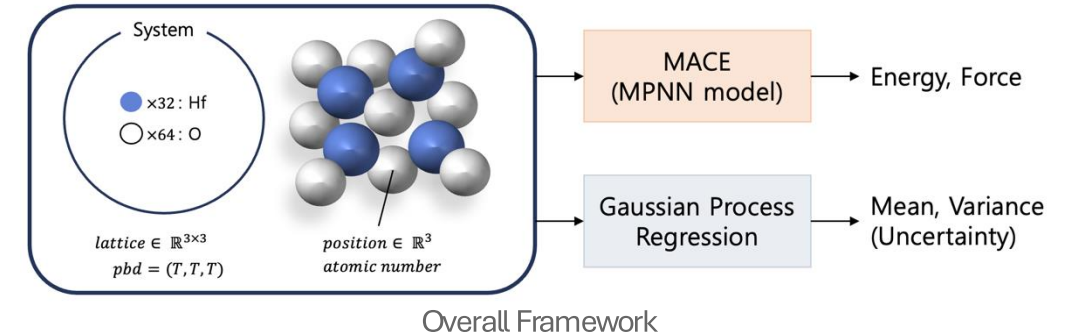
- Developed a machine learning force field for molecular energy and force prediction, along with a Gaussian Process Regression-based framework for OOD uncertainty estimation.

My Contributions

- Trained and optimized a MACE-based message passing neural network for molecular property prediction.
- Improved model performance through hyperparameter optimization, ensemble learning, and architectural exploration.
- Evaluated architectural modifications, including Product Layer, Readout Layer, Residual Connection, DropEdge, and Dropout.
- Implemented Gaussian Process Regression for out-of-distribution uncertainty estimation.

Results

- Reduced the EF Score from 5.56 to 2.66.
- Achieved 2nd place on the public leaderboard



Overview of the MACE-based molecular property prediction framework.

Image Aesthetics Assessment & Captioning

Undergraduate Researcher | Multi-modal LLM Research

Project

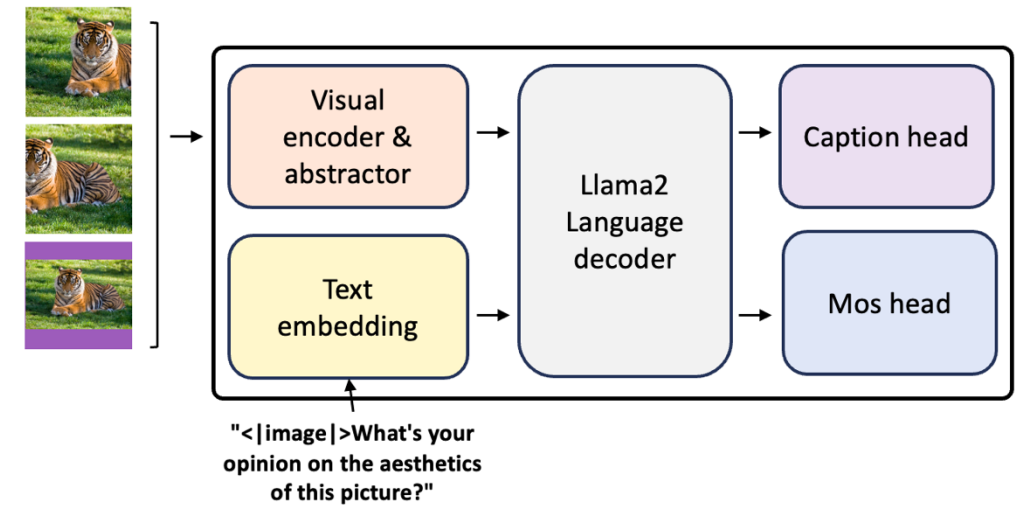
- Developed a Q-ALIGN-based multi-task multimodal LLM for joint aesthetic score prediction and caption generation, enabling both quantitative and natural-language assessment of image aesthetics.

My Contributions

- Designed a Q-ALIGN-based multi-task MLLM for joint aesthetic score prediction and caption generation.
- Implemented LLaMA2 hidden-state-based MOS prediction, autoregressive captioning, and LoRA fine-tuning.
- Improved performance through hidden-state aggregation, hyperparameter optimization, and beam-search decoding.
- Explored high-resolution visual representations, geometric order learning, and CLIP-based caption dataset purification.

Results

- Achieved SRCC 0.783 / PLCC 0.782 for aesthetic score prediction.
- Improved over Q-ALIGN LoRA (0.776 / 0.776) and VILA (0.774 / 0.774).
- Improved captioning performance through beam search and CLIP-based dataset purification.



Overall Framework

Vision-based Safety Equipment Inspection

KIST, 2023

Project

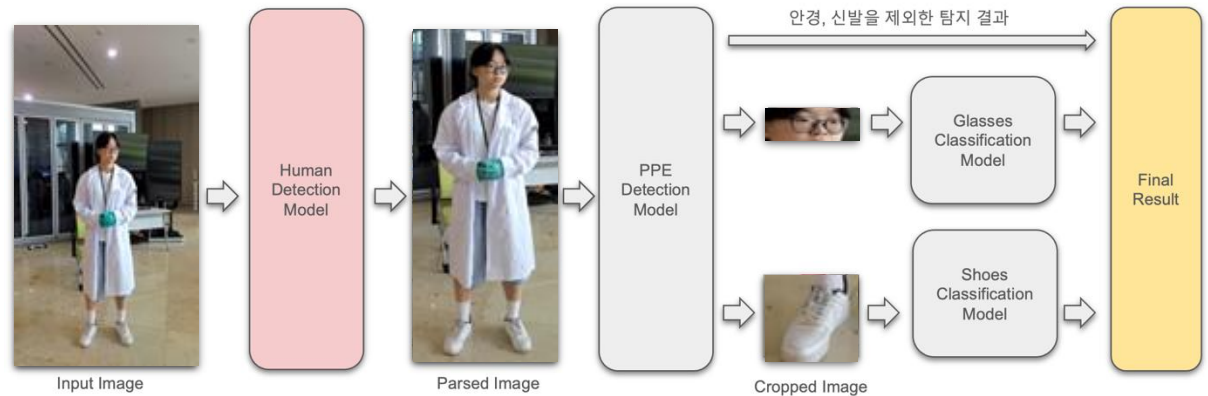
- Developed a vision-based laboratory safety inspection system with real-time PPE detection and access control.

My Contributions

- Built a 17K-image dataset and data collection pipeline.
- Enhanced detection performance through augmentation and model optimization.
- Developed a multi-stage vision pipeline for PPE inspection.
- Optimized and deployed a real-time inference system with TensorRT.

Results

- mAP: 92.2 → 98.0 (+5.8)
- Latency: 71 ms → 19 ms (3.7× Faster)
- Classification: 96.0 → 98.8



Multi-stage pipeline for laboratory safety equipment inspection.

Block / Code	Original pipeline	TensorRT applied pipeline
Human Detection	7 ms	6 ms
PPE Detection	13 ms	9 ms
Classification	51 ms	4 ms
Total Inference Time	71 ms	19 ms

Inference latency before and after TensorRT optimization.

Research Summary

Research Timeline

2023 KIST
↓
2024 Samsung AI Challenge
↓
2025 DnSAD (JSID)
↓
2026 Trajectory-aware Blur Augmentation (ICLR Target)

Technical Expertise

- Image Restoration
- Data Augmentation
- Computational Photography
- Computer Vision

Technical Expertise

Deep Learning

PyTorch • CUDA • Distributed Training

Model Development

CNN • Transformer • GNN • Vision-Language Models

Optimization & Deployment

TensorRT • Hyperparameter Optimization • Ensemble Learning

Development

Python • Docker • Linux • Git

Selected Achievements

- ICLR 2027 (Target)
- JSID 2026
- IEEE SPL 2025
- Samsung AI Challenge
- 2nd Place